

OM ANANT WAIKAR

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Education

Binghamton University, State University of New York

Expected May 2027

Master of Science in Computer Science — GPA: 3.96/4.0

Binghamton, United States

Coursework: Machine Learning, Introduction to Artificial Intelligence, Visual Information Processing, Design and Analysis of Algorithms

Indian Institute of Information Technology Vadodara

May 2025

B.Tech in Information Technology

Gujarat, India

Technical Skills

Languages: Python, Java, C, JavaScript, Dart, SQL

ML & NLP: PyTorch, Hugging Face Transformers, DistilBERT, TF-IDF, Logistic Regression

Model Evaluation: Hyperparameter Tuning, Per-Class Error Analysis, Weights & Biases

LLMs & RAG: Retrieval-Augmented Generation, LangChain, Claude API, FAISS, Document Chunking

Deployment & Tools: FastAPI, REST APIs, Docker, AWS EC2, Git, Linux/Unix

Algorithms: Graph Theory, Heuristic Algorithms, Constraint Modeling

Additional Engineering: Flutter, Realm

Experience

Catalyze Systems Pvt Ltd

Jan 2025 – Jun 2025

Flutter Developer Intern

Remote

- Shipped **15+ Flutter/Dart screens**, including Add New Patient Visit, contributing to a **50–70% increase in active user engagement**
- Integrated **10+ RESTful APIs** to exchange patient data between the mobile frontend and backend services
- Designed and managed **15+ interrelated Realm database schemas** for offline-first patient-record storage
- Brought desktop **prescription management** to mobile, supporting custom medicine entries, dosage details, and on-device PDF generation

Indian Institute of Information Technology Vadodara

Jun 2024 – Jul 2024

Summer Research Intern

Gandhinagar, India

- Implemented **3 routing heuristics** based on distance and time-window opening/closing for electric-vehicle routing with time windows (EV-TSPTW) on a **6-node graph**
- Formulated a graph-based model with a travel-distance objective and **3 constraint categories** across a **15-edge network**
- Benchmarked **3 heuristics** on distance, waiting time, and feasibility: distance-based routing minimized travel, while time-window-closing achieved the highest feasibility at a higher distance cost

Projects

Evidence-Grounded Document Q&A System | *Python, LangChain, Claude API, FAISS*

- Built a RAG pipeline that indexed 50–100 documents, retrieved relevant passages, and generated Claude-powered answers with supporting source references
- Evaluated 75–100 questions across multiple chunking and retrieval settings, reaching 80–87% answer accuracy and 85–92% citation accuracy
- Deployed the pipeline behind a **FastAPI** backend, achieving sub-second retrieval latency for interactive querying

Support-Ticket Intent Classification | *Python, PyTorch, Hugging Face Transformers, Weights & Biases*

- Fine-tuned DistilBERT on 13,000+ Banking77 examples covering 77 intent categories and compared its performance with a **TF-IDF and logistic-regression baseline**
- Tracked 15–20 training runs with approximately 87% macro-F1, using hyperparameter tuning and per-class error analysis to improve weaker categories
- Containerized the trained model with **Docker** and exposed it through a REST endpoint for real-time intent prediction

Leadership & Activities

Student Assistant, Office of International Students, Binghamton University

Feb 2026 – Present

- Supported **100+ international students weekly** through front-desk check-ins, inquiry handling, and referrals to university services
- Processed student e-forms and documentation in the **Sunapsis portal**, maintaining records and handling service requests